

Performance Analytics Kimia Farma Business Year 2020-2023

Kimia Farma - Big Data Analytics

Presented by
Sahara

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Data Analyst

I am an Industrial Engineering graduate from Malikussaleh University with a diverse professional background, including experience as a Management Trainee at PT Mitra Bisnis Keluarga Ventura and a Production Planning intern at PT Pupuk Iskandar Muda. Certified as a Data Analyst by Growia Education, I am proficient in Excel, Python, SQL, Tableau, and Google Data Studio. I specialize in strategic thinking and structured problem-solving, leveraging data-driven insights to provide high-level solutions for complex business challenges



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About Company



Established in 1817 by the Dutch East Indies Government, Kimia Farma stands as Indonesia's inaugural pharmaceutical company. Now a key member of the State-Owned Enterprise (BUMN) Pharmaceutical Holding, the company has evolved into a fully Integrated Healthcare Provider, managing an end-to-end ecosystem that includes:

- **Pharmaceutical Manufacturing:** Operating advanced production facilities for finished medicines, active pharmaceutical ingredients (APIs), and herbal products.
- **Distribution & Logistics:** Leveraging an extensive nationwide network to ensure the seamless availability and supply of healthcare products across the Indonesian archipelago.
- **Pharmaceutical Retail:** Managing the largest pharmacy chain in Indonesia, providing the public with direct access to essential medications and healthcare needs.
- **Healthcare Services:** Offering a comprehensive range of clinical laboratories and health clinics located strategically throughout the country.

Project Portfolio

This project involves a comprehensive business performance evaluation of Kimia Farma for the period of 2020 to 2023. The analysis was conducted using a robust dataset of 672,000 rows and 16 columns, integrated from various sources including Final Transactions, Product catalogs, Inventory records, and Branch Office data. By synthesizing these data points, the project provides a data-driven overview of the company's operational efficiency and sales trends across its extensive network.

[Link To GitHub](#)

1. Importing Dataset to BigQuery

This analysis was conducted utilizing **BigQuery** and **Looker Studio** as the primary analytical tools.

A dedicated project titled "**Rakamin_KF_Analytics**" was established in BigQuery, housing a dataset named "**kimia_farma**."

The analysis integrated the following core datasets:

- kf_final_transaction.csv: Comprehensive sales and transaction records.
- kf_inventory.csv: Stock levels and inventory management data.
- kf_kantor_cabang.csv: Details and locations of branch offices.
- kf_product.csv: Product master data and pricing information.

Select a resource

No organization ▼

New project

▼	Kimia_Farma	☆	⋮
	kf_final_transact	☆	⋮
	kf_inventory	☆	⋮
	kf_kantor_caban	☆	⋮
	kf_product	☆	⋮

▶ rakamin-kf-analytics-482512 ☆ ⋮

SQL query

Create dataset

Refresh contents

Upload to project ▶

1. Importing Dataset to BigQuery

```
CREATE TABLE Kimia_Farma.final_task_dataset AS
SELECT
  x.transaction_id, x.date,x.branch_id, x.branch_name,x.kota,x.provinsi, x.rating_cabang, x.customer_name, x.product_id,
  x.product_name, x.actual_price,x.discount_percentage,
  x.persentase_gross_laba, x.nett_sales,
  -- Calculation for Nett Profit: (Actual Price * Gross Laba %) - (Actual Price - Nett Sales)
  (x.actual_price * x.persentase_gross_laba) - (x.actual_price - x.nett_sales) AS nett_profit,
  x.rating AS rating_transaksi,
FROM (
  SELECT
    a.transaction_id, a.date,a.branch_id, b.branch_name, b.kota,b.provinsi, b.rating AS rating_cabang, a.customer_name,
    a.product_id, c.product_name, c.price AS actual_price,
    a.discount_percentage,
    -- Profit percentage logic based on price tiers
    CASE
      WHEN c.price <= 50000 THEN 0.10
      WHEN c.price > 50000 AND c.price <= 100000 THEN 0.15
      WHEN c.price > 100000 AND c.price <= 300000 THEN 0.20
      WHEN c.price > 300000 AND c.price <= 500000 THEN 0.25
      WHEN c.price > 500000 THEN 0.30
    END AS persentase_gross_laba,
    -- Calculation for Nett Sales: Price after discount
    (c.price - (c.price * a.discount_percentage)) AS nett_sales,
    a.rating
  FROM `Kimia_Farma.kf_final_transaction` AS a
  LEFT JOIN `Kimia_Farma.kf_kantor_cabang` AS b
    ON a.branch_id = b.branch_id
  LEFT JOIN `Kimia_Farma.kf_product` AS c
    ON a.product_id = c.product_id
) x;
```

[Link To Query](#)

2. Tabel Analisa

Adapun Tabel Analisa yang dihasilkan terdiri dari 16 kolom dan 672.458 baris sebagai berikut:

☐	Field name	Type	Mode	Description	Key	Collation	Default Value	Policy Tags [?]	Data Policies
☐	transaction_id	STRING	NULLABLE	-	-	-	-	-	-
☐	date	DATE	NULLABLE	-	-	-	-	-	-
☐	branch_id	INTEGER	NULLABLE	-	-	-	-	-	-
☐	branch_name	STRING	NULLABLE	-	-	-	-	-	-
☐	kota	STRING	NULLABLE	-	-	-	-	-	-
☐	provinsi	STRING	NULLABLE	-	-	-	-	-	-
☐	rating_cabang	FLOAT	NULLABLE	-	-	-	-	-	-
☐	customer_name	STRING	NULLABLE	-	-	-	-	-	-
☐	product_id	STRING	NULLABLE	-	-	-	-	-	-
☐	product_name	STRING	NULLABLE	-	-	-	-	-	-
☐	price	INTEGER	NULLABLE	-	-	-	-	-	-
☐	discount_percentage	FLOAT	NULLABLE	-	-	-	-	-	-
☐	persentase_gross_laba	FLOAT	NULLABLE	-	-	-	-	-	-
☐	rating_transaction	FLOAT	NULLABLE	-	-	-	-	-	-
☐	nett_sales	FLOAT	NULLABLE	-	-	-	-	-	-
☐	nett_profit	FLOAT	NULLABLE	-	-	-	-	-	-

2. Tabel Analisa

Job information		Results	Visualization	JSON	Execution details	Execution graph			
Row	transaction_id	date	branch_id	branch_name	kota	provinsi	rating_cabang	customer_name	product_id
1	TRX6546352	2021-02-23	96065	Kimia Farma - Apotek	Karawang	Jawa Barat	5.0	Chelsea Williams	KF172
2	TRX4605325	2022-05-09	99182	Kimia Farma - Apotek	Makassar	Sulawesi Selatan	4.6	Keith Shaffer	KF172
3	TRX6445517	2023-11-18	80557	Kimia Farma - Apotek	Pariaman	Sumatera Barat	4.0	Stephen Marshall	KF172
4	TRX2616052	2022-05-22	31604	Kimia Farma - Apotek	Pontianak	Kalimantan Barat	4.4	Tiffany Berry	KF172
5	TRX2619435	2022-10-17	59275	Kimia Farma - Apotek	Garut	Jawa Barat	4.6	William Anderson	KF172
6	TRX6802081	2020-08-20	34489	Kimia Farma - Apotek	Garut	Jawa Barat	4.4	Paul Rivera	KF172
7	TRX8383972	2020-07-22	64080	Kimia Farma - Apotek	Batam	Kepulauan Riau	4.1	Luis Smith	KF172
8	TRX5651355	2023-12-26	14433	Kimia Farma - Apotek	Gorontalo	Gorontalo	4.0	James Patterson	KF172
9	TRX3719390	2020-05-24	63815	Kimia Farma - Apotek	Sukabumi	Jawa Barat	4.7	Elizabeth Gutierrez	KF172
10	TRX2927649	2023-04-17	97760	Kimia Farma - Apotek	Cikampek	Jawa Barat	4.6	Tyler Weaver	KF172
11	TRX5046417	2022-03-22	42820	Kimia Farma - Apotek	Mataram	Nusa Tenggara Barat	4.1	Andrew Bates	KF172
12	TRX1232342	2023-05-30	62312	Kimia Farma - Apotek	Cianjur	Jawa Barat	4.5	Linda Butler	KF172
13	TDV2566479	2022-11-24	94899	Kimia Farma - Apotek	Tedimkulau	Jawa Barat	4.1	Susan Bradford	KF172

product_id	product_name	actual_price	discount_percent	persentase_gros	nett_sales	nett_profit	rating_transaksi
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	-105.0	3.1
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	-105.0	3.3
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	-105.0	3.2
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	-105.0	3.7
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	-105.0	3.8
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	-105.0	4.2
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	-105.0	5.0
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	-105.0	3.2
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15	0.1	1785.0	-105.0	4.6
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15				
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15				
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15				
KF172	Psycholeptics drugs, Hypnotics ...	2100	0.15				

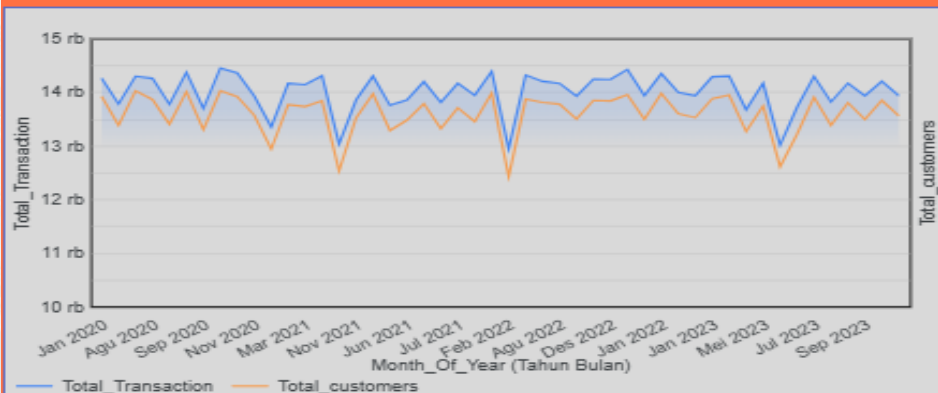
[Link To Sheet](#)

3. BigQuery Syntax

--Total Transaction and total Customer per month (Time Frame 2020-2023)

```
SELECT
  DATE_TRUNC(DATE(date), MONTH) AS Month_Of_Year,
  COUNT(DISTINCT transaction_id) AS Total_Transaction,
  COUNT(DISTINCT customer_name) AS Total_customers
FROM
  `rakamin-kf-analytics-482512.Kimia_Farma.final_task_dataset`
GROUP BY
  1
ORDER BY
  1;
```

Total Transaction and Total Customer per Month



Row	Month_Of_Year	Total_Transaction	Total_customers
1	2020-01-01	14259	13133
2	2020-02-01	13346	12347
3	2020-03-01	14366	13204
4	2020-04-01	13772	12719
5	2020-05-01	14447	13220
6	2020-06-01	13778	12705
7	2020-07-01	14161	13011
8	2020-08-01	14253	13088
9	2020-09-01	13692	12638
10	2020-10-01	14293	13217

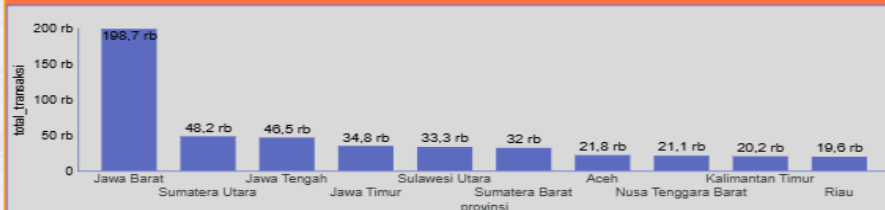
[Link To Sheet](#)

[Link To Query](#)

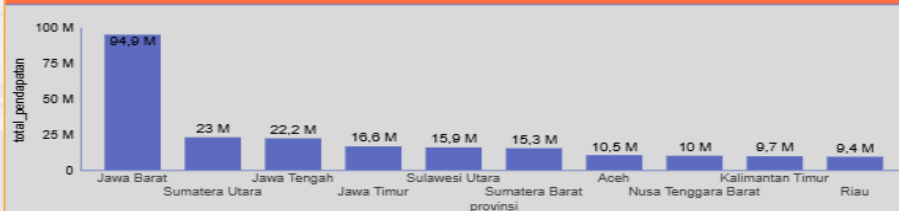
3. BigQuery Syntax

```
--TOP 10 Total transaksi dan Nett Sales Cabang Provinsi
SELECT
    provinsi,
    COUNT(DISTINCT transaction_id) AS total_transaksi,
    ROUND(SUM(netto_sales)) AS total_pendapatan
FROM
    `rakamin-kf-analytics-
482512.Kimia_Farma.final_task_dataset`
GROUP BY
    1
ORDER BY
    total_transaksi DESC
LIMIT 10;
```

Top 10 Transaksi Cabang Provinsi



Top 10 Pendapatan Cabang Provinsi



Row	provinsi	total_transaksi	total_pendapatan
1	Jawa Barat	198723	94869594875.0
2	Sumatera Utara	48178	22952159897.0
3	Jawa Tengah	46494	22248101144.0
4	Jawa Timur	34766	16627080704.0
5	Sulawesi Utara	33339	15902759535.0
6	Sumatera Barat	32014	15314406695.0
7	Aceh	21825	10467971437.0
8	Nusa Tenggara Barat	21069	10036738806.0
9	Kalimantan Timur	20243	9714536616.0
10	Riau	19607	9354616403.0

[Link To Query](#)

3. BigQuery Syntax

-- Top 5 Cabang Dengan Rating Tertinggi, namun Rating Transaksi Terendah

```
SELECT
  branch_name,
  kota,
  provinsi,
  AVG(rating_cabang) AS avg_rating_cabang,
  AVG(rating_transaksi) AS avg_rating_transaksi
FROM `rakamin-kf-analytics-
482512.Kimia_Farma.final_task_dataset`
GROUP BY 1, 2, 3
ORDER BY
  avg_rating_cabang DESC, -- Prioritaskan rating cabang
  tertinggi
  avg_rating_transaksi ASC -- Kemudian pilih rating
  transaksi terendah
LIMIT 5;
```

[Link To Query](#)

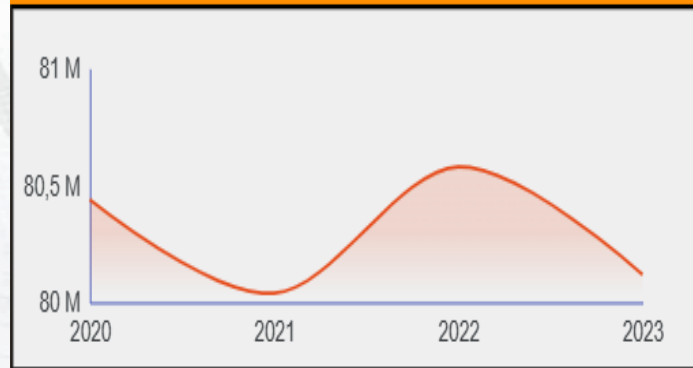
Row	branch_name	kota	provinsi	avg_rating_cabang	avg_rating_transa...
1	Kimia Farma - Klinik-Aptek-Laboratorium	Pematangsiantar	Sumatera Utara	4.830083333333...	3.991749999999...
2	Kimia Farma - Klinik & Apotek	Jambi	Jambi	4.771483375959...	4.002941176470...
3	Kimia Farma - Klinik-Aptek-Laboratorium	Batam	Kepulauan Riau	4.728635204081...	3.998596938775...
4	Kimia Farma - Apotek	Sorong	Papua Barat	4.725298804780...	3.993758300132...
5	Kimia Farma - Klinik-Aptek-Laboratorium	Cikampek	Jawa Barat	4.717094370860...	4.020198675496...

3. BigQuery Syntax

```
--Perbandingan Pendapatan Kimia Farma dari tahun ke tahun
SELECT
  EXTRACT(YEAR FROM DATE(date)) AS year,
  ROUND(SUM(net sales), 2) AS total_revenue,
FROM
  `rakamin-kf-analytics-482512.Kimia_Farma.final_task_dataset`
GROUP BY
  year
ORDER BY
  year;
```

Row	year	total_revenue
1	2020	80437605040.0
2	2021	80037846824.0
3	2022	80578445844.0
4	2023	80117292611.0

Pendapatan KF PerTahun



[Link To Query](#)

4. Dashboard Performance Analytics



[Link To
Dashboard](#)

5. Insight & Analisis

- Terdapat **1.725** cabang yang tersebar di seluruh Kota di Indonesia. Dengan Total Transaksi sebanyak **672.458**, **264.601** pelanggan dan pendapatan bersih sebesar **72,6 M** selama 2020 hingga 2023
- Dilihat dari Analisa yang dilakukan pendapatan Kimia Farma menunjukkan stabilitas yang sangat tinggi dengan pendapatan bersih (*nett sales*) konsisten di angka **Rp80 Miliar per tahun** dan **4,5 M** per Quartal
- Provinsi Jawa Barat merupakan kontributor terbesar baik dari sisi volume transaksi (**198,7 ribu**) maupun pendapatan bersih (**Rp94,9 Miliar**).
- Urutan Total Transaksi dan Nett Sales Cabang Provinsi hampir sama. Dilihat dari Top 10 Transaksi dan Nett Sales yang ada, diketahui bahwa penjualan tersebar diseluruh Indonesia, bukan hanya terpusat di satu pulau.
- Kategori obat-obatan ***Psycholeptics drugs (Hypnotics & Anxiolytics)*** menjadi kontributor penjualan tertinggi dengan total penjualan **134rb item**, total profit mencapai lebih dari **15 M** dan rata rata profit per item **114 rb** dengan persentase penjualan sebesar **20,1%** dari seluruh produk yang ada.

6. Strategi & Rekomendasi

- Dengan total 264.601 pelanggan dan 672.458 transaksi (rata-rata ~2,5 transaksi per pelanggan dalam 3 tahun), perusahaan bisa mendorong frekuensi kunjungan melalui promo rutin bulanan atau layanan cek kesehatan gratis.
- Identifikasi cabang dari total 1.725 cabang yang memiliki performa di bawah rata-rata. Lakukan evaluasi apakah perlu dilakukan relokasi atau perubahan strategi pemasaran lokal.
- Tingkatkan efisiensi rantai pasok (*supply chain*) ke cabang-cabang di luar Pulau Jawa untuk menekan biaya logistik, sehingga margin keuntungan di wilayah tersebut bisa mendekati performa di Jawa Barat. Gunakan aplikasi pemesanan online untuk menjangkau daerah yang jauh dari Lokasi cabang di berbagai daerah.

Thank You



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Academy

